

RUBY, AK, USA

WMO: 702627

Lat: 64.727N

Lon: 155.470W

Elev: 653

StdP: 14.35

Time Zone: -9.00 (NAK)

Period: 06-19

WBAN: 26561

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.8	-31.2	-34.2	0.8	-28.1	-31.2	1.0	-24.9	29.4	13.1	26.3	9.3	2.4	80	0.708

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.9	78.2	60.0	74.3	58.0	70.9	56.9	62.0	74.2	59.9	70.5	58.3	68.0	5.8	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.9	70.8	64.5	55.3	66.6	63.0	53.9	63.3	61.8	27.9	74.1	26.5	70.6	25.4	68.0	67.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.3	19.4	17.0	-37.2	84.0	6.8	3.1	-42.1	86.2	-46.1	88.0	-49.9	89.7	-54.9	92.0		
(5)			-30.7	64.1	2.7	2.3	-32.6	65.8	-34.2	67.1	-35.7	68.4	-37.7	70.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.7	-3.8	5.8	11.7	30.3	47.6	58.2	60.0	54.1	43.8	26.8	6.9	1.0
	DBStd	25.33	18.32	13.63	12.56	10.83	9.78	6.24	6.35	5.57	7.48	9.30	11.95	14.18
	HDD50	8608	1668	1237	1189	590	157	6	3	22	207	718	1292	1519
	HDD65	13310	2133	1657	1654	1040	540	220	183	339	635	1183	1742	1984
	CDD50	818	0	0	0	0	84	251	311	150	22	0	0	0
	CDD65	47	0	0	0	0	2	16	27	2	0	0	0	0
	CDH74	353	0	0	0	0	26	114	206	7	0	0	0	0
	CDH80	55	0	0	0	0	1	21	33	0	0	0	0	0
Wind	WSAvg	6.1	6.7	9.0	7.1	7.1	5.2	4.7	4.5	4.9	6.1	6.1	6.2	6.4
Precipitation	PrecAvg	19.2	1.2	1.1	0.8	0.7	0.9	1.7	2.5	3.0	2.3	1.8	1.6	1.5
	PrecMax	23.8	3.3	3.3	2.8	1.8	1.7	2.9	4.4	4.7	5.0	5.4	5.1	3.7
	PrecMin	13.1	0.2	0.0	0.0	0.1	0.1	0.4	0.8	1.4	0.8	0.3	0.1	0.2
	PrecStd	3.1	0.8	0.7	0.6	0.5	0.4	0.7	0.9	0.9	1.2	1.0	1.1	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.2	33.1	40.5	57.8	77.8	83.2	84.0	74.6	65.4	48.1	34.8	31.5
		MCWB	33.3	29.7	34.0	44.3	55.7	61.4	62.5	60.4	53.5	40.6	31.1	29.5
	2%	DB	29.7	30.6	36.6	53.7	72.6	77.0	79.6	70.3	61.2	43.3	30.2	27.8
		MCWB	28.3	28.2	30.6	41.4	53.7	58.6	61.0	58.4	51.0	40.0	29.0	26.4
	5%	DB	25.6	27.5	32.9	49.9	66.8	73.5	75.6	66.4	58.0	40.8	26.0	23.5
		MCWB	24.4	24.8	28.1	39.4	51.0	56.8	60.3	56.5	49.7	37.9	24.6	22.2
	10%	DB	20.1	23.9	29.2	45.7	62.5	69.9	71.5	63.6	54.7	38.2	22.0	20.0
		MCWB	18.8	21.9	25.3	37.3	48.7	55.4	58.1	55.2	48.4	35.9	20.9	18.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.8	30.7	36.0	44.9	56.8	64.0	65.1	62.6	54.1	43.1	32.1	30.2
		MCDB	35.4	32.0	40.2	56.8	74.8	78.3	79.8	71.9	63.0	44.9	33.5	31.6
	2%	WB	28.4	28.4	31.3	42.0	54.4	60.7	62.7	59.7	52.3	40.7	29.1	26.6
		MCDB	29.6	30.5	35.7	52.0	69.9	73.1	76.4	66.7	59.1	42.9	30.6	28.0
	5%	WB	24.9	25.1	28.2	39.9	52.2	58.5	60.5	58.0	50.6	38.2	24.8	22.4
		MCDB	26.2	27.4	32.6	48.8	65.1	70.3	72.3	64.6	56.2	40.2	26.2	23.5
	10%	WB	19.9	21.9	25.2	37.5	50.1	56.7	59.2	56.4	49.2	36.2	21.0	19.0
		MCDB	21.0	23.7	29.2	45.1	60.3	67.7	69.8	62.2	54.5	38.0	22.0	20.0
Mean Daily Temperature Range	5% DB	MDBR	9.7	11.2	15.6	16.3	17.1	16.4	14.9	13.1	11.6	7.7	8.8	8.8
		MCDBR	10.1	11.3	16.1	18.9	22.0	20.2	20.2	17.4	16.7	9.3	10.1	8.8
		MCWBR	9.1	9.9	11.9	10.6	10.0	8.7	8.7	8.2	9.2	6.9	9.6	8.1
	5% WB	MCDBR	10.4	11.8	15.3	17.8	20.3	18.0	18.2	14.3	14.1	8.2	9.9	8.8
		MCWBR	9.4	10.5	11.5	10.3	9.3	8.6	8.4	7.8	8.3	6.5	9.5	8.2
Clear-Sky Solar Irradiance	taub		0.170	0.245	0.250	0.314	0.332	0.365	0.367	0.365	0.303	0.264	N/A	0.174
	taud		1.939	2.205	2.302	2.216	2.434	2.358	2.397	2.409	2.559	2.461	N/A	1.930
	Ebn at Noon		125	215	272	271	279	269	265	253	252	213	N/A	40
	Edh at Noon		7	17	24	34	30	34	32	29	20	14	N/A	3
All-Sky Solar Radiation	RadAvg		54	266	838	1462	1682	1740	1485	1113	708	287	80	15
	RadStd		6	25	64	102	167	131	113	103	93	36	12	1

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)			+1.94	N/A	+1.53	N/A	N/A	N/A	-574	-691	+113

Nomenclature: See separate page